

Multipoles, Macroscopic Media, Dielectrics

PHYS505 – Electromagnetic Theory

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Multipole expansion

- General solution for the potential in spherical coordinates

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \left[A_{lm} r^l + B_{lm} r^{-(l+1)} \right] Y_{lm}(\theta, \phi)$$

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- With $r = r_>$, $r' = r_<$

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{(r')^l}{r^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$

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$$q_{lm} = \int_{r' < R} Y_{lm}^*(\theta', \phi') (r')^l \rho(\vec{x}') d^3x'$$

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$$q_{00} = \frac{1}{\sqrt{4\pi}} \int \rho(\vec{x}') d^3x' = \frac{q}{\sqrt{4\pi}}$$

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$$q_{lm} = \int_{r' < R} Y_{lm}^*(\theta', \phi') (r')^l \rho(\vec{x}') d^3x'$$

$$q_{11} = -\sqrt{\frac{3}{8\pi}} \int (x' - iy') \rho(\vec{x}') d^3x' = -\sqrt{\frac{3}{8\pi}} (p_x - ip_y)$$

$$q_{10} = \sqrt{\frac{3}{4\pi}} \int z' \rho(\vec{x}') d^3x' = \sqrt{\frac{3}{4\pi}} p_z$$

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$$q_{lm} = \int_{r' < R} Y_{lm}^*(\theta', \phi') (r')^l \rho(\vec{x}') d^3x'$$

$$\begin{aligned} q_{22} &= \frac{1}{4} \sqrt{\frac{15}{2\pi}} \int (x' - iy')^2 \rho(\vec{x}') d^3x' \\ &= \frac{1}{12} \sqrt{\frac{15}{2\pi}} (Q_{11} - 2iQ_{12} - Q_{22}) \end{aligned}$$

$$q_{21} = -\sqrt{\frac{15}{8\pi}} \int z'(x' - iy') \rho(\vec{x}') d^3x' = \frac{1}{12} \sqrt{\frac{15}{2\pi}} (Q_{13} - iQ_{23})$$

$$q_{20} = \frac{1}{2} \sqrt{\frac{5}{4\pi}} \int (3z'^2 - r'^2) \rho(\vec{x}') d^3x' = \frac{1}{2} \sqrt{\frac{5}{4\pi}} Q_{33}$$

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$$q_{l,-m} = (-1)^m q_{lm}$$

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$$q_{lm} = \int_{r' < R} Y_{lm}^*(\theta', \phi') (r')^l \rho(\vec{x}') d^3x'$$

- $\vec{p}$ is the electric dipole moment (vector)

$$\vec{p} = \int \vec{x}' \rho(\vec{x}') d^3x'$$

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$$q_{lm} = \int_{r' < R} Y_{lm}^*(\theta', \phi') (r')^l \rho(\vec{x}') d^3x'$$

- Q_{ij} is the (traceless) quadrupole moment tensor

$$Q_{ij} = \int (3x'_i x'_j - r'^2 \delta_{ij}) \rho(\vec{x}') d^3x'$$

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- In rectangular coordinates,

$$\Phi = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r} + \frac{\vec{p} \cdot \vec{x}}{r^3} + \frac{1}{2} \sum_{i,j} Q_{ij} \frac{x_i x_j}{r^5} + \dots \right]$$

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$$\Phi(\vec{x}) = \frac{1}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{q_{lm}}{2l+1} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}}$$

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- Electric field

$$\vec{E} = -\vec{\nabla}\Phi$$

$$E_r = -\partial_r \Phi = \frac{1}{\epsilon_0} \sum_{l,m} \frac{l+1}{2l+1} \frac{q_{lm}}{r^{l+2}} Y_{lm}(\theta, \phi)$$

$$E_\theta = -\frac{1}{r} \partial_\theta \Phi = -\frac{1}{\epsilon_0} \sum_{l,m} \frac{1}{2l+1} \frac{q_{lm}}{r^{l+2}} \partial_\theta Y_{lm}(\theta, \phi)$$

$$E_\phi = -\frac{1}{r \sin \theta} \partial_\phi \Phi = \frac{1}{\epsilon_0} \sum_{l,m} \frac{1}{2l+1} \frac{q_{lm}}{r^{l+2}} \frac{im}{\sin \theta} Y_{lm}(\theta, \phi)$$

Multipole expansion

$$\Phi(\vec{X}) = \frac{1}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{q_{lm}}{2l+1} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}}$$

- Dipole along z

$$E_r = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3} \cos \theta$$

$$E_\theta = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \sin \theta$$

$$E_\phi = 0$$

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- More generally, for a dipole at $\vec{x}_0$,

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \left[\frac{3\hat{n}(\vec{p} \cdot \hat{n}) - \vec{p}}{|\vec{x} - \vec{x}_0|^2} - \frac{4\pi}{3} \vec{p} \delta(\vec{x} - \vec{x}_0) \right]$$

Multipole expansion

- Moments in general depend on the choice of origin
- Point charge e at $\vec{x} = \vec{x}_0$

$$\rho(\vec{r}) = e \delta(\vec{x} - \vec{x}_0)$$

$$q_{lm} = e \int_{r < R} Y_{lm}^*(\theta, \phi) r^l \delta(\vec{x} - \vec{x}_0) d^3x = e Y_{lm}^*(\theta_0, \phi_0) r_0^l$$

- If $\vec{x}_0 = \vec{0}$,

$$q_{lm} = \begin{cases} e Y_{00} = \frac{e}{\sqrt{4\pi}}, & l = 0 \\ 0, & l > 0 \end{cases}$$
$$\Phi = \frac{1}{\epsilon_0} q_{00} \frac{Y_{00}}{r} = \frac{1}{\epsilon_0} \frac{e}{\sqrt{4\pi}} \frac{1}{\sqrt{4\pi} r}$$

- If $\vec{x}_0 \neq \vec{0}$, q_{00} is the same but other $q_{lm} \neq 0$

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- $q_{lm} \neq 0$ for $l > 1$ unless $\vec{x}_0 = -\vec{x}_1 = \pm z_0 \hat{z}$
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Charge distribution in external field

- Localized $\rho(\vec{x})$ in external potential $\Phi(\vec{x})$, energy of the system

$$W = \int \rho(\vec{x}) \Phi(\vec{x}) d^3x$$

- If Φ varying slowly over the region

$$\begin{aligned}\Phi(\vec{x}) &= \Phi(\vec{0}) + \vec{x} \cdot \vec{\nabla} \Phi(\vec{0}) + \frac{1}{2} \sum_{i,j} x_i x_j \partial_{ij}^2 \Phi(\vec{0}) + \dots \\ &= \Phi(\vec{0}) - \vec{x} \cdot \vec{E}(\vec{0}) - \frac{1}{6} \sum_{i,j} (3x_i x_j - r^2 \delta_{ij}) \partial_i E_j(0) + \dots\end{aligned}$$

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- States $|JM_\alpha\rangle$ for nucleus with quadrupole moment (cylindrically symmetric) ρ_{JM_α}

$$\frac{1}{e} Q_{33} = Q_{JM_\alpha} = \int (3z^2 - r^2) \rho_{JM_\alpha}(\vec{x}) d^3x$$

- causes splitting of energy levels in external (crystal lattice or molecular) field
- Interaction energy between two dipoles

$$W_{12} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p}_1 \cdot \vec{p}_2 - 3(\hat{n} \cdot \vec{p}_1)(\hat{n} \cdot \vec{p}_2)}{|\vec{x}_1 - \vec{x}_2|^2}$$

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Charge distribution in external field

$$W = q\Phi(\vec{0}) - \vec{p} \cdot \vec{E}(\vec{0}) - \frac{1}{6} \sum_{i,j} Q_{ij} \partial_i \partial_j \Phi(\vec{0}) + \dots$$

- States $|JM\alpha\rangle$ for nucleus with quadrupole moment (cylindrically symmetric) $\rho_{JM\alpha}$

$$\frac{1}{e} Q_{33} = Q_{JM\alpha} = \int (3z^2 - r^2) \rho_{JM\alpha}(\vec{x}) d^3x$$

- causes splitting of energy levels in external (crystal lattice or molecular) field
- Interaction energy between two dipoles

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- and didn't need to distinguish between microscopic and macroscopic (average) fields
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Dielectric media

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$$\vec{P}(\vec{x}) = \sum_i N_i \langle \vec{p}_i \rangle$$

for material made up of mixture of density N_i of molecules of type i with dipole moment $\vec{p}_i$ with the average over a (macroscopically) small volume around $\vec{x}$

- If the molecules have net charge e_i and there's excess (macroscopic) charge density in the problem

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- Contribution to potential from induced dipole moments in material

$$\Delta\Phi = \frac{1}{4\pi\epsilon_0} \left[\frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} + \frac{\vec{P}(\vec{x}') \cdot (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \right] \Delta V$$

$$\begin{aligned}\Phi(\vec{x}) &= \frac{1}{4\pi\epsilon_0} \int d^3x' \left[\frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} + \vec{P}(\vec{x}') \cdot \vec{\nabla}' \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) \right] \\ &= \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{|\vec{x} - \vec{x}'|} [\rho(\vec{x}') - \vec{\nabla}' \cdot \vec{P}(\vec{x}')] \end{aligned}$$

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- Defining the *electric displacement* $\vec{D}$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}, \quad \vec{\nabla} \cdot \vec{D} = \rho$$

- For a linear isotropic medium

$$\vec{P} = \chi_e \epsilon_0 \vec{E}, \quad \vec{D} = \epsilon \vec{E}, \quad \epsilon = (1 + \chi_e) \epsilon_0$$

where

- χ_e is called the *electric susceptibility*
- ϵ/ϵ_0 is the *dielectric constant* or *relative permittivity*
- If the medium is also uniform

$$\vec{\nabla} \epsilon = \vec{0} \rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon}$$

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Dielectric interfaces

At a boundary between two dielectric media

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{n}_{21} = \sigma$$

$$(\vec{E}_2 - \vec{E}_1) \times \hat{n}_{21} = 0$$

where

- $\hat{n}_{21}$ is a unit surface normal pointing from 1 to 2
- σ is the excess macroscopic charge on the surface

BVP with dielectrics – point charge near interface

- Point charge q in semi- ∞ medium with ϵ_1 distance d away from interface $z = 0$ with ϵ_2

BVP with dielectrics – point charge near interface

- $\epsilon_2][\epsilon_1 \leftarrow d \rightarrow q \Rightarrow \hat{\mathbf{z}}$
- $\vec{E}$ satisfies

$$\vec{\nabla} \times \vec{E} = 0 \rightarrow \vec{E} = -\vec{\nabla}\Phi$$

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$$\vec{\nabla} \cdot \vec{D} = \epsilon_1 \vec{\nabla} \cdot \vec{E} = q \delta(\vec{x} - d\hat{\mathbf{z}}), \quad z > 0$$

BVP with dielectrics – point charge near interface

- $\epsilon_2 \left[\epsilon_1 \leftarrow d \rightarrow q \Rightarrow \hat{\mathbf{z}} \right.$
- $\text{bc } (E_\rho, \epsilon_2 E_z)(z \rightarrow 0^-) = (E_\rho, \epsilon_1 E_z)(z \rightarrow 0^+)$

BVP with dielectrics – point charge near interface

- $q' \leftarrow d \rightarrow \epsilon_2$ [$\epsilon_1 \leftarrow d \rightarrow q \Rightarrow \hat{\mathbf{z}}$]
- bc $(E_\rho, \epsilon_2 E_z)(z \rightarrow 0^-) = (E_\rho, \epsilon_1 E_z)(z \rightarrow 0^+)$
- Try image charge q' at $z = -d$

$$\Phi_1(z > 0) = \frac{1}{4\pi\epsilon_1} \left(\frac{q}{R_1} + \frac{q'}{R_2} \right) \quad \nabla^2 \Phi_1 = q \delta(\vec{x} - d\hat{\mathbf{z}})$$
$$R_1 = \sqrt{\rho^2 + (d - z)^2} \quad R_2 = \sqrt{\rho^2 + (d + z)^2}$$

BVP with dielectrics – point charge near interface

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$$= \frac{q''}{\epsilon_2}$$

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BVP with dielectrics – point charge near interface

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BVP with dielectrics – point charge near interface

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- Image charges q' at $z = -d$ and q'' at $z = d$

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$$2q = \left(1 + \frac{\epsilon_1}{\epsilon_2}\right) q''$$

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$$\frac{q''}{q} = \frac{2}{1 + \frac{\epsilon_1}{\epsilon_2}} = \frac{2\epsilon_2}{\epsilon_2 + \epsilon_1}$$

BVP with dielectrics – point charge near interface

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- Image charges q' at $z = -d$ and q'' at $z = d$

$$-(q - q' = q'')$$

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$$2q' = \left(-1 + \frac{\epsilon_1}{\epsilon_2}\right) q''$$

$$\frac{q'}{q''} = \frac{1}{2} \left(\frac{\epsilon_1}{\epsilon_2} - 1\right) = \frac{\epsilon_1 - \epsilon_2}{2\epsilon_2}$$

BVP with dielectrics – point charge near interface

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$$\frac{q'}{q''} = \frac{\epsilon_1 - \epsilon_2}{2\epsilon_2}$$

$$\frac{q'}{q} = \frac{q'}{q''} \frac{q''}{q} = \frac{\epsilon_1 - \epsilon_2}{\epsilon_1 + \epsilon_2}$$

BVP with dielectrics – point charge near interface

- $q' \leftarrow d \rightarrow \epsilon_2$ || $\epsilon_1 \leftarrow d \rightarrow q, q'' \Rightarrow \hat{\mathbf{z}}$
- bc $(E_\rho, \epsilon_2 E_z)(z \rightarrow 0^-) = (E_\rho, \epsilon_1 E_z)(z \rightarrow 0^+)$

$$E_z|_{0^-} = -\frac{q'}{4\pi\epsilon_2} \frac{d}{(\rho^2 + d^2)^{3/2}}, \quad E_z|_{0^+} = -\frac{q''}{4\pi\epsilon_1} \frac{d}{(\rho^2 + d^2)^{3/2}}$$

- Image charges q' at $z = -d$ and q'' at $z = d$

$$q' = \frac{\epsilon_1 - \epsilon_2}{\epsilon_1 + \epsilon_2} q, \quad q'' = \frac{2\epsilon_2}{\epsilon_2 + \epsilon_1} q$$

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$$\sigma_{pol} = (\vec{P}_2 - \vec{P}_1) \cdot \hat{\mathbf{z}}$$

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- For $\epsilon_2 \gg \epsilon_1$, left side looks like a conductor:

$$E(z < 0) \ll E(z > 0), \quad \sigma \approx \epsilon_0 E_z|_{0^+}$$

Molecular polarizability and electric susceptibility

$$\vec{P} = \chi_e \epsilon_0 \vec{E}$$

- $\vec{E}$ is the macroscopic field
- Field $\vec{E}_{near}$ near an individual molecule includes the *internal* field $\vec{E}_i$ due to its neighbors in addition to the macroscopic field $\vec{E}_P$ which is the average response of the medium

$$\vec{E}_i = \vec{E}_{near} - \vec{E}_P$$

- Within a spherical region of radius R around the molecule,

$$\vec{p} = \frac{4\pi}{3} R^3 \vec{P}$$

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Molecular polarizability

- In a harmonic molecular binding model, charges displaced by $\vec{x}$ due to external $\vec{E}$

$$m\omega_0^2\vec{x} = e\vec{E}$$

$$\vec{p}_{mol} = e\vec{x} = \frac{e^2}{m\omega_0^2}\vec{E}$$

$$\gamma_{mol} = \frac{1}{\epsilon_0} \sum_k \frac{e_k^2}{m_k \omega_k^2}$$

for a set of charges e_k with mass m_k and frequency ω_k in each molecule.

- With ω corresponding to optical frequencies

$$\lambda \approx 0.3 \mu\text{m}, \quad c \simeq 0.3 \text{ mm THz}, \quad \omega \approx 6000 \text{ THz}$$

- order of magnitude estimate for electronic γ

$$\gamma_{el} = \frac{e^2}{\epsilon_0 m_e \omega^2} \approx 10^{-29} \text{ m}^3 \approx (\text{few } a_0)^3$$

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- Harmonic binding with external E_z

$$H = \frac{1}{2m}\vec{p}^2 + \frac{1}{2}m\omega_0^2 \vec{x}^2 - e E z$$

- Accounting for thermal (Boltzmann) distribution doesn't change p_{mol}

$$f(H) = e^{-H/kT}, \quad \langle p_{mol} \rangle = \frac{e^2}{m\omega_0^2} E$$

- For polar substances with permanent dipole moment $\vec{p}_0$, polarizability reduced by temperature as thermal fluctuations disturb alignment with external field

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Electrostatic energy in dielectric media

- Energy of a charge distribution assembled by bringing in pieces against the field of the rest

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Electrostatic energy in dielectric media

- System with initial field $\vec{E}_0$ from fixed sources

$$W_0 = \frac{1}{2} \int \vec{E}_0 \cdot \vec{D}_0 d^3x$$

- Adding object with linear response ($\epsilon = \epsilon_1$) changes the field to $\vec{E}$

$$W_1 = \frac{1}{2} \int \vec{E} \cdot \vec{D}_0 d^3x$$

- The difference can be written as

$$W = W_1 - W_0 = -\frac{1}{2} \int_{V_1} (\epsilon_1 - \epsilon_0) \vec{E} \cdot \vec{E}_0 d^3x = -\frac{1}{2} \int_{V_1} \vec{P} \cdot \vec{E}_0 d^3x$$

where V_1 is the dielectric region

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- Energy density within object placed in vacuum with field from fixed sources

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- Compare to energy from multipole expansion

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$$w = -\frac{1}{2} \vec{P} \cdot \vec{E}_0$$

- Compare to energy from multipole expansion

$$W = q\Phi_0 - \vec{p} \cdot \vec{E}_0 + \dots$$

for a permanent dipole in an external field $\vec{E}_0$

- For $\epsilon_1 > \epsilon_0$, the force on the object will be toward higher E_0